

Raspberry Pi 4 Model B (Core Brain of the System)

What it is

A powerful single-board computer with:

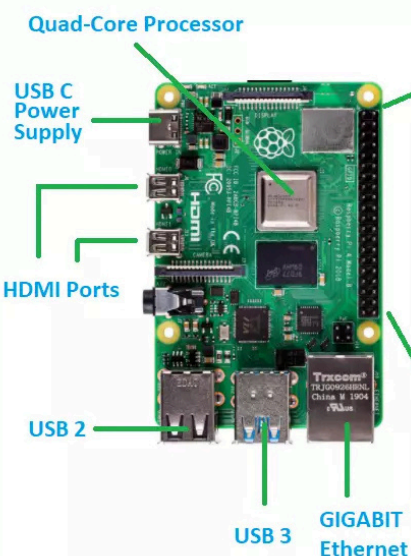
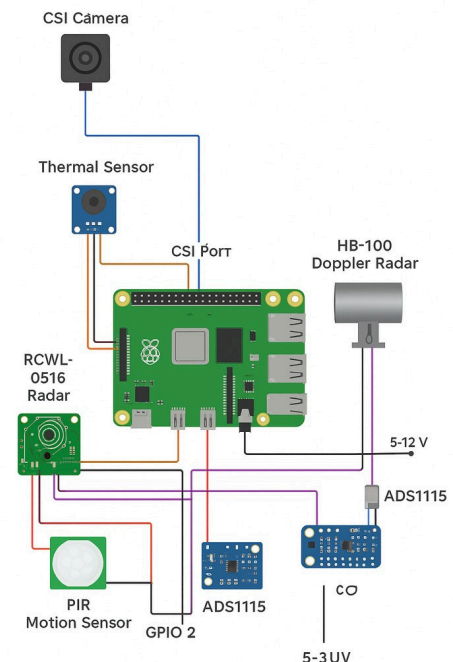
- Quad-core ARM Cortex-A72 CPU (1.5 GHz)
- 2/4/8GB RAM
- CSI Camera connector
- GPIO pins for sensors
- Runs full Linux OS

Why it is used

- Runs **YOLOv8** human detection
- Processes **thermal images**
- Reads **radar + PIR + auxiliary sensors**
- Generates **heatmaps**
- Hosts **dashboard server (Flask/WebRTC)**
- Triggers **alerts & logs**

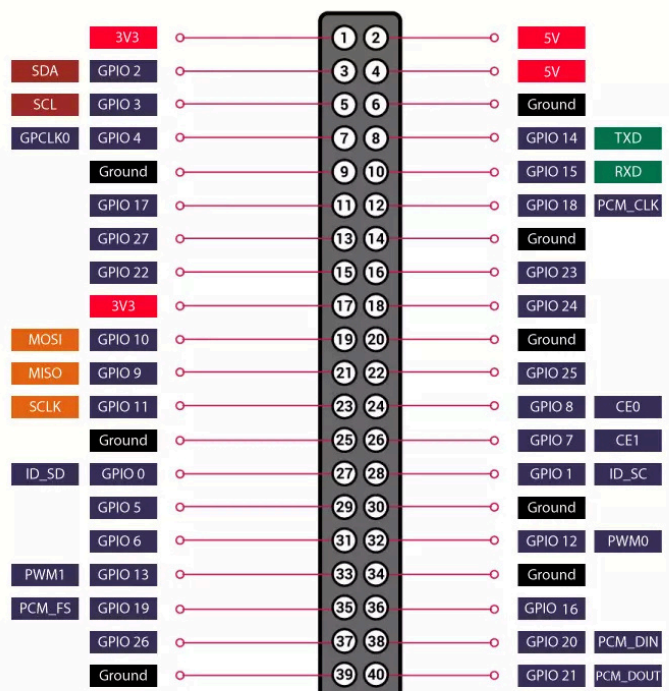
Ports Used

- **CSI Port** → Camera **I²C Pins** → Thermal
- sensors **GPIO** → Radar, PIR, buzzers
- **USB** → Optional WiFi dongles, peripherals



Raspberry Pi 4 Pinout

www.theengineeringprojects.com



Raspberry Pi Camera (CSI Camera / V2 / HQ Camera)

📌 Function

Captures visible-light video frames for:
YOLOv8 human detection
Live monitoring dashboard
Heatmap overlay (thermal fusion)

📌 Why it is important

YOLO needs RGB image frames
CSI camera gives low-latency, high FPS, unlike USB webcams
Works with:
OpenCV
Libcamera
GStreamer

📌 How it works

Connects directly to CSI port
Sends raw frames to Pi GPU at up to 120 FPS
YOLO processes frames to detect:
Humans
Body posture
Multiple persons

📌 CSI Wiring

Just plug in ribbon cable → Camera.

📌 Outputs

640×480 or 1280×720 feed
Bounding boxes + confidence
Frame buffer for dashboard

OV5647 5MP Camera – Hardware Overview

The OV5647 is a 5-megapixel CMOS image sensor designed by OmniVision.

It is widely used in Raspberry Pi modules because it is cheap, reliable, and optimized for embedded systems.

✓ Sensor Type

CMOS digital camera sensor
Bayer filter
Rolling shutter

✓ Resolution

5 MP (2592 × 1944 pixels)

✓ Video Support

1080p at 30 FPS
720p at 60 FPS
QVGA at 120 FPS

✓ Interface

MIPI CSI-2 (2-lane interface) → Direct connection to Raspberry Pi camera port

✓ Lens Type

Fixed focus or adjustable manual focus (in some variants)

✓ Field of View

Standard version: ~54°

Wide-angle versions: 120°+, depending on lens

How the OV5647 Camera Works Internally (Simplified)

- ✓ Step 1 – Light hits camera lens
Matches onto Bayer filter (RGB color space).
- ✓ Step 2 – Photodiodes convert light → electrons
Every pixel stores charge proportional to light intensity.
- ✓ Step 3 – CMOS sensor converts electrons → digital pixel value
Rolling shutter reads pixels row by row.
- ✓ Step 4 – Image goes into the ISP (Image Signal Processor)

The ISP does:

- Debayering
- Noise reduction
- Auto exposure
- Auto white balance
- Auto gain control
- Lens shading corrections
- ✓ Step 5 – MIPI CSI-2 interface sends image to Raspberry Pi GPU
CSI is very fast, giving:
 - High bandwidth
 - Low latency (great for real-time YOLO detection)
- ✓ Step 6 – OpenCV/libcamera captures frames
These frames are passed to:
 - YOLO (AI detection)
 - Dashboard streaming (MJPEG/WebRTC)

4. OV5647 Camera Modes & Resolutions

Still Image Resolution

- 2592×1944 (5 MP)

Video Modes

Resolution	FPS
1080p	30 FPS
720p	60 FPS
VGA	90 FPS
QVGA	120 FPS

High FPS modes are great for motion detection.

The OV5647 5MP camera is used for:

- ✓ Visible light human detection
Detects body shape, posture, clothing, movement
- ✓ Supports multi-sensor fusion
This combines with:
 - Thermal sensor
 - Radar sensor
 - PIR sensor
- ✓ Provides:
 - Live video stream
 - Visual confirmation
 - Evidence capture
 - Threat classification

Advantages of OV5647 Camera

- ✓ Highly optimized for Raspberry Pi
Designed to work directly with the GPU ISP.
- ✓ Low power
Only 250–300 mW.
- ✓ Very low latency
Important for real-time detection.
- ✓ Cheap
₹250–₹500 in India.
- ✓ Many lens options
 - Standard
 - Wide-angle
 - Telephoto
 - Night-vision with IR LEDs

1. AMG8833 Specifications (Important to Know)

Feature	Details
Resolution	8×8 (64 pixels)
Temperature Range	0°C to 80°C
Human Detection Range	0–7 meters
Accuracy	±2.5°C
Thermal Refresh Rate	10 FPS max
Interface	I ² C
Supply Voltage	3.3V ONLY
View Angle	~60° × 60°
Sensor Type	Infrared Array (Thermopile-based)

Because it uses **thermopile sensors**, it can detect heat without contact.

2. AMG8833 Pinout Explained Simply

The breakout board has 5 pins:

AMG8833 Pin	Function
VIN	3.3V power
GND	Ground
SDA	I ² C data
SCL	I ² C clock
INT	Interrupt (optional, usually unused)

Important:

- ✓ Always use 3.3V, NEVER 5V directly (unless module has regulator).
- ✓ Uses I²C address 0x69 or 0x68 (depends on model).

3. AMG8833 → Raspberry Pi 4 Wiring Diagram

Basic Wiring

AMG8833 Pin	Raspberry Pi 4 Pin
VIN	3.3V (Pin 1 or 17)
GND	GND (Pin 6, 9, 14, etc.)
SCL	GPIO 3 (SCL – Pin 5)
SDA	GPIO 2 (SDA – Pin 3)
INT (optional)	Any GPIO, e.g., GPIO 4

This is the **correct and safest connection**.

How the AMG8833 Works (Internal Working)

✓ Step 1 – IR Heat Detection

Each of the 64 pixels has a thermopile IR sensor.

These detect IR radiation emitted by warm bodies (humans emit ~9–10 μm wavelength).

✓ Step 2 – Sensor converts IR → temperature

Each pixel outputs a temperature value between 0°C and ~80°C.

Humans typically appear between 30°–37°C.

✓ Step 3 – Chip sends temperature values over I²C

The Pi reads 64 temperature values at up to 10 frames/second.

✓ Step 4 – Pi interpolates 8×8 → heatmap

Using algorithms like:

bicubic interpolation

bilinear interpolation

the Pi converts low-res 8×8 data → smooth 64×64 or 240×240 heatmap.

✓ Step 5 – Human detection logic

If pixel cluster has:

temperature > threshold

contiguous hot region (3×3 or 4×4)

→ Human detected

- ✓ Human detection using thermal signature
People show up as bright clusters.
- ✓ Night surveillance
Works perfectly in zero light.
- ✓ Smoke / fog detection
Thermal does NOT depend on visible light.
- ✓ Combined fusion with YOLO camera
YOLO sees the human
Thermal confirms heat signature
Radar confirms movement
= Highly accurate detection with low false positives
- ✓ Intruder detection
Sudden rise of heat pixels → alert
Good for:
Doors
Corridors
Disaster rescue
- ✓ Fire detection
Detecting temperature spikes > 60°C.

Advantages of AMG8833

- Small & inexpensive
- Low power
- Works in total darkness
- Gives real heat data
- Easy I²C interface
- Perfect for edge-AI

Limitations

- VERY low resolution (8×8 = 64 pixels)
- Cannot recognize faces
- Cannot detect exact shape
- Not good beyond 7 meters
- Slow refresh rate (10 FPS)
- Needs interpolation for visuals

Heatmap Visualization (Dashboard Use)

- You can generate:
 - Thermal image (color-coded)
 - Hotspot map
 - Human-shaped blob
 - Thermal bounding box
- AMG8833 is ideal for:
 - Heat signature detection
 - Long-distance presence sensing
 - Using in dark or smoke environments

MLX90640 (32x24 Thermal IR Array)

High-resolution thermal camera for Raspberry Pi
768 pixels (32 × 24)
Best for human detection, heatmaps, night vision, disaster rescue.

1. MLX90640 Overview / What It Is

The Melexis MLX90640 is a non-contact thermal imaging sensor that measures IR radiation from all 768 pixels simultaneously.

Key Specifications

Parameter	Value
Resolution	32 × 24 (768 pixels)
Temperature Range	-40°C to 300°C
Human Detection Range	0–15 meters
FPS	1–16 Hz (depending on mode)
Interface	I ² C (up to 1MHz)
Operating Voltage	3.3V only
FOV Options	55°×35° (standard), 110°×75° (wide)
Accuracy	±2°C

Compared to AMG8833 (8x8), this is 4 times higher resolution → sharper heatmaps.

2. MLX90640 Pin Diagram & Connections to Raspberry Pi 4

MLX90640 Pinout (Typically 6-pin board)

Pin	Function	Description
VIN	3.3V	Power input (Some modules support 5V, but prefer 3.3V)
GND	Ground	Common ground
SCL	I ² C Clock	Connect to Pi SCL
SDA	I ² C Data	Connect to Pi SDA
INT (optional)	Interrupt	Not required for basic usage
AD0 (optional)	Address select	Sets I ² C address (0x33 or 0x32)

3. MLX90640 → Raspberry Pi 4 Wiring

Simple 4-wire connection (recommended)

MLX90640 Pin	Raspberry Pi Pin
VIN	3.3V (Pin 1 or 17)
GND	GND (Pin 6 or 9)
SCL	GPIO 3 (Pin 5)
SDA	GPIO 2 (Pin 3)

Optional:

MLX90640 Pin	Pi
INT	Any GPIO (ex: GPIO 4)
AD0	Leave unconnected or set HIGH/LOW

Internal Working of MLX90640 (in easy terms)

- ✓ Step 1 – IR radiation enters the microlens array
Humans emit infrared radiation around 9–10 μm wavelength.
- ✓ Step 2 – IR hits the thermopile array
Each of the 768 thermopiles converts heat $\rightarrow$ voltage.
- ✓ Step 3 – Analog front-end amplifies signals
MLX90640 has:
 - Chopper-stabilized amplifiers
 - Noise reduction
 - Pixel offset compensation
- ✓ Step 4 – On-chip DSP calculates:
 - Pixel temperature
 - Compensation values
 - Ambient temperature
 - Calibration data
- ✓ Step 5 – Final temperature frame sent via I²C
Raspberry Pi reads:
 - 32×24 = 768 pixel temperatures
- ✓ Step 6 – Heatmap generated
Pi interpolates $\rightarrow$ smooth 240×320 or 480×640 heatmap.
This creates a true thermal image.

Usage in SIH Human Detection Project

- ✓ Detect humans in total darkness
MLX works without light.
- ✓ Detect hidden humans
Behind:
 - Smoke
 - Fog
 - Light barriers
- ✓ Distance detection (0–10 meters)
Humans show up as:
 - 32×24 pixel cluster
 - Temperature gradient difference

✓ Fusion with YOLO camera

Your system combines:

Sensor	What it Provides
MLX90640	Heat signature
Pi Camera (OV5647)	Visual YOLO detection
RCWL Radar	Motion detection
PIR Sensor	Warm-body presence

Fusion increases accuracy to 95%+.

✓ Detect Fire

Spots regions > 80°C.

MLX90640 Heatmap Visualization

Process:

- Get 768 pixel temperatures
- Interpolate to larger resolution
- Normalize temperature values
- Apply color map (JET, Inferno, VIRIDIS)
- Overlay on camera feed (if needed)
- Heatmap shows:
 - Humans as bright blobs
 - Fire as extremely bright region
 - Cold objects as blue

HB-100 Microwave Doppler Radar Sensor

The HB-100 is a 10.525 GHz microwave Doppler radar module used to detect motion, speed, and direction of moving objects — including humans, vehicles, and intruders.

It works through walls, plastic, wood, fog, smoke, making it very powerful.

What Exactly Is HB-100?

It is a continuous-wave microwave Doppler motion sensor.

✓ Emits 10.525 GHz microwaves

The sensor sends microwaves into the environment.

✓ Receives reflected waves

When objects move, the reflected wave frequency changes.

✓ Calculates Doppler Shift

This frequency change is directly proportional to movement speed.

✓ Outputs analog IF (intermediate frequency) voltage signal

This is why Raspberry Pi cannot read it directly — it needs an ADC.

2. HB-100 Technical Specifications

Parameter	Value
Frequency	10.525 GHz
Detection Range	5–20 meters
Type	Doppler radar
Output	Analog IF signal
Supply Voltage	5V
Current	30–40 mA
Detects	Motion, speed, direction
Works through	Walls, plastic, wood, glass

3. HB-100 Pin Diagram (Important)

Typical HB-100 modules have 4 pins:

Pin	Function	Description
VCC	5V	Power the microwave transmitter/receiver
GND	Ground	Common ground
IF	Intermediate Frequency Output	Analog signal representing motion/speed
OSC	Oscillator output (optional)	Not used typically

The important pin:

👉 IF Pin = Doppler output waveform

(This must go into ADS1115 ADC)

Why HB-100 CANNOT be connected directly to Raspberry Pi

Raspberry Pi does NOT have analog input pins, only digital GPIOs.

But HB-100 outputs:

Small analog voltages (millivolts)

Doppler frequency (Hz-to-kHz range)

So if you connect IF → GPIO, Pi will not read anything.

✓ YOU MUST USE AN ADC:

Most common:

ADS1115 (recommended)

16-bit ADC

High sensitivity

Reads tiny HB-100 signals

5. Wiring HB-100 → ADS1115 → Raspberry Pi 4

Below is the correct wiring.

✓ HB-100 → ADS1115 Wiring

HB-100 Pin	ADS1115 Pin	Description
VCC	5V external supply	HB-100 requires clean 5V
GND	GND	Common ground
IF	A0	Read analog Doppler signal

(You can also use A1/A2/A3)

✓ ADS1115 → Raspberry Pi 4 Wiring

ADS1115 Pin	Raspberry Pi Pin
VDD	3.3V
GND	GND
SDA	GPIO 2 (Pin 3)
SCL	GPIO 3 (Pin 5)
ADDR	GND (sets I ² C address 0x48)

✓ 5V Power Note

HB-100 must run on 5V because its microwave oscillator needs correct power.

Pi 4's 5V pin is clean enough:

- Pin 2 = 5V
- Pin 4 = 5V

Advantages of HB-100

- ✓ Long range (up to 20 meters)
Better than PIR/ultrasonic.
- ✓ Works through walls
Unlike most sensors.
- ✓ Detects speed
This helps threat classification.
- ✓ Immune to lighting
Night, fog, smoke, dust — works same.
- ✓ Very low power

Limitations

- ✗ Cannot detect stationary humans
Use PIR + Thermal for that.
- ✗ Metal walls block microwaves
Physics limitation.
- ✗ Needs ADC
Cannot connect directly to Pi.
- ✗ Sensitive to vibrations

HB-100 helps:

Detect movement early
Trigger camera + thermal scanning
Provide speed for threat score
Detect movement behind walls
It is part of your multi-layer detection pipeline.

How HB-100 Works Internally (Simple Explanation)

✓ Step 1 – Microwave Transmission

The HB-100 emits 10.525 GHz microwaves constantly.

✓ Step 2 – Object Movement

If an object (human/vehicle) moves:

Waves reflect back slightly shifted in frequency.

✓ Step 3 – Doppler Shift

The change in frequency (Δf) is:

$$\Delta f = (2 * V) / \lambda$$

Where:

V = speed of object

λ = wavelength (~2.85 cm)

✓ Step 4 – Sensor outputs analog frequency

Slow motion → low frequency (0–50 Hz)

Walking human → 20–200 Hz

Running → 200–500 Hz

Vehicle → 1–3 kHz

✓ Step 5 – Raspberry Pi reads IF values via ADC

The ADC captures:

Waveform

Frequency

Amplitude changes

7. What HB-100 Can Detect in Your Project

✓ Human Movement Detection

Even if:

Camera cannot see

Thermal blocked

Darkness

Smoke or fog

Behind non-metal walls

✓ Speed of Human Movement

You can detect:

Walking

Running

Stopping

Direction changes

✓ Distance Estimation

Waveform amplitude gives approximate range.

✓ Security / Surveillance

Continuous motion scanning for:

Intruders

Suspicious movement

Fast movements alert

RCWL-0516 Microwave Radar Sensor –

The RCWL-0516 is a 5.8 GHz microwave Doppler radar module, used as a motion detection sensor. It detects movement by sending microwave waves and analyzing the reflected patterns. It is far more sensitive than PIR sensors and works:

- Through walls
- In complete darkness
- Through plastic/wood/glass
- Over several meters

What is RCWL-0516?

It is a microwave motion detection module that replaces PIR sensors for more accurate and far-range movement detection.

- ✓ Key Capabilities
 - Detects moving humans reliably
 - Works in 360° around itself
 - Detects motion through doors, plastic, thin walls
 - Immune to:
 - Heat changes
 - Dust
 - Fog
 - Darkness
 - Fast response (sub-100 ms)
 - Low power

2. RCWL-0516 Technical Specifications

Parameter	Value
Frequency	5.8 GHz
Voltage	4–28V (works perfectly on 5V)
Current	2–3 mA
Output	Digital HIGH/LOW
Detection Distance	5–7 meters
Detection Type	Motion only (via Doppler)
Output Hold Time	~2 seconds
Output Voltage	3.3V logic
Angle	360° (spherical detection)

3. RCWL-0516 Pin Diagram

The module has 5 pins:

Pin	Function	Description
VIN	4–28V	Power input (use 5V from Pi)
GND	Ground	Common ground
OUT	Output	Digital signal (HIGH = motion detected)
CDS	Disable mode (optional)	For light-based activation (rarely used)
3V3	Regulated 3.3V out	Not for powering module

→ You only need 3 pins: VIN, GND, OUT

4. RCWL-0516 → Raspberry Pi 4 Wiring

✓ Only 3 wires needed

RCWL-0516 Pin	Raspberry Pi Pin
VIN	5V (Pin 2 or 4)
GND	GND (Pin 6, 9, 14 etc.)
OUT	GPIO 17 (Pin 11)

🔧 Output logic

- Motion detected → OUT = 3.3V (HIGH)
- No motion → LOW

This means it is directly compatible with Pi’s GPIO (3.3V safe).

How RCWL-0516 Works (Simple Explained)

The sensor uses Doppler shift of microwave waves.

✓ Step 1 – Sensor emits 5.8 GHz microwaves

The waves spread in all directions.

✓ Step 2 – Waves bounce off objects

Stationary objects → constant reflection

Moving objects → shifted frequency reflection

✓ Step 3 – Frequency difference analyzed

If Δf > threshold → sensor detects motion.

✓ Step 4 – Sensor drives output HIGH

🎯 This detects:

Walking human

Running human

Hand movement

Hidden movement behind thin walls/doors

It does NOT detect:

Standing still person

Heat

Temperature changes

(That's what PIR and Thermal sensors are for.)

6. Why RCWL-0516 is Better Than PIR Sensors

Feature	PIR	RCWL-0516
Detects through walls	✗	✓
Range	3–5m	7+ m
Sensitivity	Low	Very high
Temperature dependency	High	None
Night-time sensitivity	Reduced	Same
360° field	✗	✓

RCWL-0516 is a true microwave radar, not IR-based.

The RCWL-0516 contributes to early motion detection.

✓ 1. Pre-Trigger for Camera & Thermal

Before YOLO & MLX90640 start heavy processing, radar detects motion.

Radar HIGH → Activate camera + thermal scan

Saves power & improves FPS.

✓ 2. Combine with Thermal Sensor

If both show activity:

Radar = HIGH

Thermal hotspot = TRUE

→ HUMAN CONFIRMED (high confidence)

✓ 3. Good for Night Surveillance

Even when camera sees nothing:

Radar detects movement

Thermal detects heat

Fusion detects human

✓ 4. Detect Hidden Intruders

Works through:

Glass

Plastic

Thin plywood

Clothing

Cardboard

✓ 5. Threat Scoring

Fast movement = higher threat

Slow movement = medium threat

No movement = low threat

10. Limitations

Limitation	Explanation
Cannot detect standing humans	Needs movement
Highly sensitive	May detect small vibrations
Triggered by fans, motors	Microwave reflections
Cannot measure distance or speed	Only motion/no motion

Waveshare Human Detection Microwave Sensor

Waveshare makes two major types of human-detection microwave radar sensors:

Waveshare 24GHz Human Presence Sensor (HPS Series)

Waveshare Doppler Radar Motion Sensor (RCWL-like)

The one used in security/surveillance projects is usually the:

★ 24GHz Human Presence Detection Radar (Waveshare HPS-24 / LD2410 Compatible)

It is far more advanced than RCWL-0516.

Key Features (Why It Is Better Than RCWL-0516)

✓ Uses 24GHz FMCW radar, not 5.8GHz CW

→ Can differentiate micro-motions (breathing, slight movement).

✓ Can detect still humans

Unlike RCWL-0516 and PIR.

✓ Detection Range

Micro-motion: 0–6 meters

Macro motion: 0–10 meters

✓ Outputs multiple types of data

Presence

Motion speed

Distance

Energy values

✓ Works through:

Plastic

Glass

Thin wood

Light non-metallic walls

✓ Immune to:

Heat

Sound

Vibrations

Light conditions

Dust, fog, smoke

✓ Low power consumption (5V, <80mA)

This makes it one of the best sensors for indoor human detection.

Sensor Variants

Waveshare’s human-detection radar modules come with:

✓ Digital Output (HIGH/LOW)

→ Easy GPIO usage

✓ UART Interface (TX/RX)

→ Distance, energy, speed, presence

✓ I2C Interface (some versions)

→ Data packet mode

The most common is the UART + GPIO model.

4. Waveshare Human Detection Radar Pin Diagram

Typical pins:

Pin	Function	Description
5V	Power input	4.5–5.5V
GND	Ground	Common ground with Pi
OUT / IO	Digital Output	HIGH = human presence
TX	UART TX	Sensor → Pi data
RX	UART RX	Pi → sensor config
EN	Enable	Pull HIGH to activate
RST	Reset	Optional

Depending on model, pins may vary slightly.

5. Connection to Raspberry Pi 4

✓ Basic Wiring (Digital Output Only)

Waveshare Pin	Raspberry Pi Pin
5V	Pin 2 or 4 (5V)
GND	Any GND pin
OUT	GPIO 17 (or any input pin)
EN (optional)	3.3V (Pin 1)

Behavior:

- OUT = HIGH → Human detected
- OUT = LOW → No human

✓ Advanced Wiring (UART mode – Recommended)

Waveshare Pin	Raspberry Pi Pin
5V	5V (Pin 2/4)
GND	GND
TX	GPIO 15 (RXD)
RX	GPIO 14 (TXD)
EN	3.3V

This allows Raspberry Pi to read:

- Distance (in cm)
- Motion energy
- Static energy
- Movement speed
- Detection zone mapping
- Presence vs motion

UART Setup on Raspberry Pi

Enable UART:

```
sudo raspi-config
```

Go to:

Interfaces → Serial → Enable Serial Port, Disable Login Shell

Install python UART library:

```
sudo pip install pyserial
```

How Waveshare Human Detection Radar Works (Internal Explanation)

This sensor uses 24 GHz FMCW (Frequency Modulated Continuous Wave) radar, the same principle used in automotive and security radars.

✓ Step 1 – Sensor emits 24GHz swept-frequency signals
Microwaves continuously sweep across a frequency band.

✓ Step 2 – Waves hit objects and reflect back
Humans reflect microwaves differently from objects.

✓ Step 3 – Micro-motion detection (unique feature)
The radar can detect:

- Chest movement during breathing
- Slight arm/leg movement
- Micro vibrations caused by human presence

✓ Step 4 – DSP analyzes return signal
Extracts:

- Doppler shift
- Micro-motion energy
- Static reflection pattern
- Distance
- Speed

✓ Step 5 – Output classification
The sensor intelligently classifies:

- Presence detected
- Static human present
- Moving human
- Human leaving
- No human

This makes it superior to PIR and RCWL sensors.

7. Output Data Explanation (UART Mode)

In UART mode, sensor outputs structured data:

Data Type	Meaning
Presence	1 = human, 0 = no human
Motion Energy	How fast object is moving
Static Energy	Body stillness detection
Distance	Human distance (0-5m)
Speed	Motion Doppler value
Detection Zone	9-zone grid

This is perfect for AI fusion systems.

- ✓ 1. Detect humans even when not moving
Thermal + Visual + RCWL cannot detect completely still humans.
Waveshare radar can detect micro-motion (breathing).
- ✓ 2. Detect humans behind thin walls
Useful for rescue, monitoring, and security.

. Advantages

- ✓ Detects still & moving humans
- ✓ Works in total darkness
- ✓ Works through obstacles
- ✓ High accuracy
- ✓ Long life, stable
- ✓ No false triggers from heat (unlike PIR)
- ✓ Better range and reliability than RCWL-0516

Limitations

- ✗ Cannot detect behind metal
- ✗ Shorter range than HB-100
- ✗ More expensive
- ✗ Needs UART parsing for full features